Numeric II - Homework 10

1) $y'' = \sin(y) + \cos(y)$

a) Let $q = y$, $p = y'$. Then
$\dot{q} = p$, $\dot{p} = y'' = \sin q + \cos q$.
The Hamiltonian equations

$$\dot{q} = \frac{\partial H}{\partial p} \quad , \quad \dot{p} = -\frac{\partial H}{\partial q}$$

$\Rightarrow \dot{q} = p = H_p$, it means

$$H(p,q) = \frac{1}{2}p^2 + V(q)$$, where V is some potential

$$\dot{p} = -H_q = \sin q + \cos q \Rightarrow H_q = -(\sin q + \cos q)$$

$$\dot{p} = -H_q = \sin q + \cos q = -\left(\frac{\partial}{\partial q}\left(\frac{1}{2}p^2 + V(q)\right)\right) = -V'(q)$$

$\Rightarrow H_q = V'(q)$. Then

$$H_q = V'(q) = -(\sin q + \cos q) \Rightarrow V(q) = \cos q - \sin q.$$

Hence, $H(p,q) = \frac{1}{2}p^2 + \cos q - \sin q$.

b) Let $p(t) = y'(t)$, $q(t) = y(t)$

$$\frac{d}{dt}H(p(t), q(t)) = H_p\,\dot{p} + H_q\,\dot{q}$$

We know $H_p = p$, $H_q = -\sin q - \cos q$. Then by using
$\dot{q} = p$, $\dot{p} = \sin q + \cos q$.

$$\Rightarrow \frac{d}{dt}H = p(\sin q + \cos q) + (-\sin q - \cos q)p = 0.$$

Therefore, $H(y'(t), y(t)) = $ constant in t.

c) $H(p,q) = \frac{1}{2}p^2 + \cos q - \sin q$ (part a)

$H(y'(t), y(t)) = \text{constant} = C$ (part b)

$\Rightarrow \frac{1}{2}(y')^2 + \cos y - \sin y = C.$

Note that $\cos y - \sin y = (\cos y, \sin y) \cdot (1,-1)$

By Cauchy-Schwarz, we have

$$|\cos y - \sin y| \leqslant \underbrace{\|(\cos y, \sin y)\|}_{=1} \underbrace{\|(1,-1)\|}_{=\sqrt{2}}.$$

$\Rightarrow |\cos y - \sin y| \leqslant \sqrt{2}$. It implies

$-\sqrt{2} \leqslant \cos y - \sin y \leqslant \sqrt{2}$.

$\frac{1}{2}(y')^2 + \underbrace{\cos y - \sin y}_{\geqslant -\sqrt{2}} = C \Rightarrow \frac{1}{2}(y')^2 = C + \sqrt{2}$

$\Rightarrow |y'(t)| = \sqrt{2(C+\sqrt{2})}$, $\forall t$.

There is no time dependency of t and it is bounded.

Therefore, $|y'(t)| = \sqrt{2(C+\sqrt{2})}$ is bounded

independently as desired.